

MARITIME CORE

ENGINEERING ROADMAP

(Architecture Baseline)

PHASE 0

DISCOVERY

Status

COMPLETED

Tujuan

- Observasi PT Dua Jangkar Indonesia
- Identifikasi bottleneck
- Mengumpulkan pain point
- Menentukan ruang lingkup sistem
- Menentukan kebutuhan bisnis

Output

✓ Pain Point

✓ Business Process

✓ Root Cause

✓ Improvement Opportunity

PHASE 1

ENGINEERING REASONING

Status

COMPLETED

Tujuan

- Menentukan arah sistem

- Membandingkan alternatif
- Menghilangkan solusi yang tidak efisien
- Menentukan filosofi database

Output

✓ Maritime Core

✓ Enterprise Architecture

✓ Business Driven

✓ Information Driven

PHASE 2

PARAMETER VALIDATION

Status

COMPLETED

Seluruh parameter divalidasi.

Parameter

✓ Zero Uncontrolled Redundancy

✓ Single Source of Truth

✓ Enterprise Ready

✓ Historical Traceability

✓ Scalability

✓ Modular Architecture

✓ Long-term Maintainability

✓ Dependency Driven

✓ Strong Referential Integrity

✓ Business Reality Mapping

Output

Blueprint dinyatakan VALID.

PHASE 3

ARCHITECTURE

Status

COMPLETED

Target

Membuat Blueprint Database.

Output

✓ Domain

✓ Entity

✓ Relationship

✓ Dependency

✓ Package

✓ Naming

✓ Datatype

✓ Constitution

Blueprint kemudian di LOCK.

PHASE 4

BASELINE IMPLEMENTATION

Status

ACTIVE

Target

Membangun baseline database.

Roadmap

PKG-001

Global Master

PKG-002

Principal Foundation

PKG-003

Vessel Foundation

PKG-004

Crew Foundation

PKG-005

Document Foundation

PKG-006

Recruitment

PKG-007

Deployment

PKG-008

Intelligence

PKG-009

System

Output

DDL Baseline

PHASE 5

DATASET

Status

WAITING

Target

Mengisi seluruh master.

Country

Currency

Language

Timezone

Rank

Document Type

Certificate

Nationality

Dan seluruh master lainnya.

PHASE 6

CORE CRUD

Status

WAITING

Target

Membangun CRUD.

Principal

Crew

Vessel

Document

Deployment

Recruitment

Monitoring

PHASE 7

BUSINESS ENGINE

Status

WAITING

Target

Workflow.

Recruitment

Approval

Deployment

Promotion

Rotation

Document Expiry

Notification

Monitoring

PHASE 8

INTELLIGENCE

Status

WAITING

Target

Business Intelligence.

Dashboard

Analytics

Readiness

Risk

Compliance

Recommendation

Forecast

PHASE 9

AUTOMATION

Status

WAITING

Target

Automation.

Scheduler

Reminder

Expiry Monitor

Background Job

Notification

Audit

PHASE 10

SECURITY

Status

WAITING

Target

Security Layer.

Authentication

Authorization

Permission

Encryption

Storage

Integrity

Audit Trail

PHASE 11

OPTIMIZATION

Status

WAITING

Target

Performance.

Normalization Audit

Redundancy Audit

Index Audit

Query Audit

Storage Audit

Scalability Audit

PHASE 12

BETA TEST

Status

WAITING

Target

Simulasi operasional.

Import

CRUD

Monitoring

Deployment

Reporting

Automation

PHASE 13

STABILIZATION

Status

WAITING

Target

Bug Fix.

Refinement.

Architecture Review.

Performance Review.

PHASE 14

RELEASE

Status

WAITING

Target

Enterprise Beta Release.

ENGINEERING RULE

Setiap PHASE mengikuti aturan.

Reasoning

↓

Approved

↓

Blueprint

↓

LOCK

↓

Compile

↓

Implementation

↓

Milestone Audit

↓

LOCK

↓

Next Phase

Tidak diperbolehkan melompat Phase.

Tidak diperbolehkan redesign setelah Blueprint LOCK tanpa Architecture Decision Record (ADR).

Seluruh implementasi wajib mengikuti Dependency Graph dan Package Roadmap.

END